



Selva Ganesh L 251801302-2025-AIDS ▾

S2**Started on** Wednesday, 26 August 2026, 8:49 PM**State** Finished**Completed on** Wednesday, 26 August 2026, 8:50 PM**Time taken** 1 min 14 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int countZeroes(int arr[], int low, int high, int n)
4  {
5      // No elements
6      if (low > high)
7          return 0;
8
9      int mid = low + (high - low) / 2;
10
11     // If mid is 0, check whether it is the first zero
12     if (arr[mid] == 0)
13     {
14         if (mid == 0 || arr[mid - 1] == 1)
15             return n - mid;
16
17         // There may be an earlier zero
18         return countZeroes(arr, low, mid - 1, n);
19     }
20
21     // arr[mid] is 1, so first zero must be on the right
22     return countZeroes(arr, mid + 1, high, n);
23 }
24
25 int main()
26 {
27     int m;
28
29     scanf("%d", &m);
30
31     int arr[m];
32
33     for (int i = 0; i < m; i++)
34     {
35         scanf("%d", &arr[i]);
36     }
37
38     int result = countZeroes(arr, 0, m - 1, m);
39
40     printf("%d\n", result);
41
42     return 0;
43 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Wednesday, 26 August 2026, 8:51 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 8:51 PM
Time taken	31 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int majorityElement(int nums[], int n)
4 {
5     int candidate = 0;
6     int count = 0;
7
8     for (int i = 0; i < n; i++)
9     {
10         if (count == 0)
11         {
12             candidate = nums[i];
13             count = 1;
14         }
15         else if (nums[i] == candidate)
16         {
17             count++;
18         }
19         else
20         {
21             count--;
22         }
23     }
24
25     return candidate;
26 }
27
28 int main()
29 {
30     int n;
31     scanf("%d", &n);
32
33     int nums[n];
34 }
```

```
35     for (int i = 0; i < n; i++)
36     {
37         scanf("%d", &nums[i]);
38     }
39
40     printf("%d\n", majorityElement(nums, n));
41
42     return 0;
43 }
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

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Grade 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findFloor(int arr[], int low, int high, int x)
4  {
5      // No valid floor
6      if (low > high)
7          return -1;
8
9      int mid = low + (high - low) / 2;
10
11     if (arr[mid] == x)
12     {
13         return arr[mid];
14     }
15
16     if (arr[mid] > x)
17     {
18         // Floor must be on the left
19         return findFloor(arr, low, mid - 1, x);
20     }
21
22     // arr[mid] < x
23     int result = findFloor(arr, mid + 1, high, x);
24
25     // If there is no larger valid element,
26     // arr[mid] is the floor
27     if (result == -1)
28         return arr[mid];
29
30     return result;
31 }
32
33 int main()
34 {
35     int n;
36     scanf("%d", &n);
37
38     int arr[n];
39
40     for (int i = 0; i < n; i++)
41     {
42         scanf("%d", &arr[i]);
43     }
44
45     int x;
46     scanf("%d", &x);
47
48     int floor = findFloor(arr, 0, n - 1, x);
49
50     printf("%d\n", floor);
51
52     return 0;
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct

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S2**Started on** Wednesday, 26 August 2026, 8:52 PM**State** Finished**Completed on** Wednesday, 26 August 2026, 8:53 PM**Time taken** 1 min 1 sec**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findPair(int arr[], int left, int right, int x)
4  {
5      // No pair found
6      if (left >= right)
7          return 0;
8
9      int sum = arr[left] + arr[right];
10
11     if (sum == x)
12     {
13         printf("%d\n", arr[left]);
14         printf("%d\n", arr[right]);
15         return 1;
16     }
17
18     if (sum < x)
19     {
20         // Need a larger sum
21         return findPair(arr, left + 1, right, x);
22     }
23     else
24     {
25         // Need a smaller sum
26         return findPair(arr, left, right - 1, x);
27     }
28 }
29
30 int main()
31 {
32     int n;
33     scanf("%d", &n);
34
35     int arr[n];
36
37     for (int i = 0; i < n; i++)
38     {
39         scanf("%d", &arr[i]);
40     }
41
42     int x;
43     scanf("%d", &x);
44
45     if (!findPair(arr, 0, n - 1, x))
46     {
47         printf("No\n");
48     }
49
50     return 0;
51 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

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S2**Started on** Wednesday, 26 August 2026, 8:54 PM**State** Finished**Completed on** Wednesday, 26 August 2026, 8:54 PM**Time taken** 26 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5	12 34 67 78 98
67 34 12 98 78	

Answer:

```

1  #include <stdio.h>
2
3  void swap(int *a, int *b)
4  {
5      int temp = *a;
6      *a = *b;
7      *b = temp;
8  }
9
10 int partition(int arr[], int low, int high)
11 {
12     int pivot = arr[high];
13     int i = low - 1;
14
15     for (int j = low; j < high; j++)
16     {
17         if (arr[j] <= pivot)
18         {
19             i++;
20             swap(&arr[i], &arr[j]);
21         }
22     }
23
24     swap(&arr[i + 1], &arr[high]);
25
26     return i + 1;
27 }
28
29 void quickSort(int arr[], int low, int high)
30 {
31     if (low < high)
32     {
33         int pivotIndex = partition(arr, low, high);
34
35         quickSort(arr, low, pivotIndex - 1);
36         quickSort(arr, pivotIndex + 1, high);
37     }
38 }
39
40 int main()
41 {
42     int n;
43     scanf("%d", &n);
44
45     int arr[n];
46
47     for (int i = 0; i < n; i++)
48     {
49         scanf("%d", &arr[i]);
50     }
51
52     quickSort(arr, 0, n - 1);

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

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